

# Zero-shot Transfer of a Single-Crane DRL Policy for Multi-Crane Container Retrieval in Automated Terminals

Anonymous<sup>a</sup>

<sup>a</sup>*University, Country*

---

## Abstract

Container terminals operate multiple yard cranes per block, yet the Container Retrieval Problem (CRP) has been studied exclusively as a single-crane problem. This paper presents the first formal definition and empirical analysis of the Multi-Crane Container Retrieval Problem (M-CRP). We investigate whether a Deep Reinforcement Learning (DRL) policy trained solely on single-crane instances transfers zero-shot to multi-crane environments via heuristic crane assignment strategies. On 140 M-CRP instances, ZoneSplit achieves gaps of 10.46% (C=2) and 10.71% (C=3) against the proposed M-CRP lower bound, reducing interference events by 99.5% compared to non-spatial strategies. These gaps are approximately one-third of those achieved by the best heuristic extended to multi-crane (M-Lin2015: 31.73% at C=2). On single-crane benchmarks (111 instances, 888 runs), ZeroShot achieves 6.03% gap versus 22.42% for Lin2015 ( $3.7\times$  worse) and 7.06% for the original pretrained model. Code and benchmarks are publicly released.

*Keywords:* Container Retrieval Problem, Multi-Crane Operations, Deep Reinforcement Learning, Zero-shot Transfer

---

## 1. Introduction

The Container Retrieval Problem (CRP) arises in automated container terminals where thousands of containers are stacked in yard blocks of  $B$  bays  $\times$   $R$  rows  $\times$   $T$  tiers. Retrieving a buried target container requires relocating blocking containers to other stacks—a process called reshuffling. The objective is to minimize total crane working time [1].

CRP research has progressed through three generations. Early heuristic methods, such as the Stack Selection Index [2] and critical-stack classification [3], use hand-crafted relocation rules. Metaheuristic approaches, including Tabu Search and GRASP [5, 6], apply general-purpose search procedures. Durasevic et al. [7] evolved priority functions via Genetic Programming. Most recently, Shin et al. [1] proposed a Deep Reinforcement Learning

---

*Email address:* `corresponding@author.email` (Anonymous)

(DRL) policy that achieves 7.8% gap against the theoretical lower bound on the standard Lee benchmark.

All existing CRP research assumes single-crane operation. Real terminals—Rotterdam, Singapore, Busan—operate two to three yard cranes per block [9]. Multi-crane operation introduces coordination constraints: cranes share a track, cannot occupy the same bay (collision avoidance), and must preserve spatial ordering. This creates three unresolved challenges:

1. **Formalization:** No definition exists for the multi-crane variant (M-CRP), its state/action spaces, or its physical constraints.
2. **Evaluation:** Without lower bounds or benchmarks, solution quality cannot be assessed.
3. **Transferability:** It is unknown whether single-crane policies can be reused in multi-crane settings.

Shin et al. [1] identified multi-crane operation as critical future work. This paper asks: *Can a single-crane DRL policy be transferred zero-shot to multi-crane environments via heuristic crane assignment?* We make five contributions:

- [C1] **Formal M-CRP definition** with interference constraints (A6, A7) and NP-hardness proof.
- [C2] **M-CRP lower bound (Theorem 3)** decomposing total cost into retrieval, relocation, and interference terms.
- [C3] **Four crane assignment strategies** for zero-shot single-to-multi-crane transfer.
- [C4] **Empirical evaluation** on 140 M-CRP instances (1,120 runs) with statistical testing.
- [C5] **Public benchmark** (140 instances, CC BY 4.0).

## 2. Related Work

Table 1 summarizes single-crane CRP performance on the Lee benchmark. Three observations motivate our transfer study. First, DRL methods [1] substantially outperform heuristics (7.8% vs 22.42–47.33%). Second, heuristic performance degrades significantly with yard scale (Lin2015: 5.55% on 1-bay vs 37.75% on 10-bay instances). Third, the DRL policy is publicly available, making it an ideal baseline for transfer.

\*Numbers from [1] Table 1; code not publicly available.

Table 1: Summary of single-crane CRP performance on Lee benchmark (gap % against lower bound). Lower is better.

| Method                      | Type                         | Instances | Mean Gap(%) |
|-----------------------------|------------------------------|-----------|-------------|
| Lin et al. [2]              | Heuristic (SSI)              | 111       | 22.42       |
| Kim et al. [3]              | Heuristic (classification)   | 111       | 44.32       |
| Leveling [1]                | Heuristic (height balancing) | 111       | 22.78       |
| Forster & Bortfeldt [5]     | Tabu Search                  | 36        | ~30–40*     |
| Cifuentes & Riff [6]        | GRASP                        | 36        | ~46*        |
| Durasevic et al. [7]        | Genetic Programming          | 111       | 51.33       |
| Shin et al. [1]             | DRL (Transformer)            | 36        | 7.8         |
| <b>ZeroShot (this work)</b> | Extracted DRL                | 111       | <b>6.03</b> |
| <b>Original Model [1]</b>   | Original DRL                 | 111       | 7.06        |

Multi-crane coordination has been studied for quay crane scheduling [11, 12] and yard crane deployment [13], but no prior work addresses retrieval sequencing within a single block—the problem central to CRP.

Zero-shot transfer has been demonstrated across problem sizes for routing [10] and general combinatorial optimization [8]. Cross-environment transfer—from single-agent to multi-agent—remains unexplored for container terminal problems.

### 3. Problem Formulation

#### 3.1. Single-Crane CRP

Following [1], a CRP instance is defined by yard dimensions  $(B, R, T)$  and  $N$  container priorities  $\{1, \dots, N\}$ . The objective minimizes total working time  $J = \sum_t (\text{travel}(a_t) + \text{handling} + \text{penalty})$  with cost parameters  $t_{\text{acc}} = 40$ ,  $t_{\text{bay}} = 3.5$ ,  $t_{\text{row}} = 1.2$ ,  $t_{\text{pd}} = 30$ .

#### 3.2. M-CRP: Formal Definition

**Definition 3.1** (M-CRP). *Given yard  $B \times R \times T$ ,  $C \geq 1$  cranes, and  $N$  containers with retrieval order  $\sigma$ , find a sequence  $\{(a_t, c_t)\}_{t=1}^{\tau_c}$  minimizing:*

$$\min \sum_{c=1}^C \sum_{t=1}^{\tau_c} (\text{travel}_c(a_t) + \text{handling}) \quad (1)$$

*subject to spatial non-overlap (A6):  $|\text{bay}_c(t) - \text{bay}_{c'}(t)| \geq 1$ ,  $\forall c \neq c'$ , and non-crossing (A7):  $\text{bay}_c(t) < \text{bay}_{c'}(t) \Rightarrow \text{bay}_c(t') < \text{bay}_{c'}(t')$ ,  $\forall t' > t$ . Assumptions A1–A5 of [1] remain unchanged.*

**Theorem 3.1** (NP-hardness). *M-CRP with  $C \geq 1$  is NP-hard (reduction from single-crane CRP [1]).*

### 3.3. M-CRP Lower Bound

**Theorem 3.2** (Theorem 3). *For any M-CRP instance with  $C$  cranes, the total working time is bounded below by:*

$$LB_{MCRP} = LB_{\text{retrieval}} + \frac{LB_{\text{relocation}}}{C} + LB_{\text{interference}} \quad (2)$$

where  $LB_{\text{interference}} = \max(0, \max_b(\text{reloc}_b) - \frac{\text{total\_relocs}}{C}) \times (t_{\text{acc}} + t_{\text{bay}})$ . At  $C = 1$ ,  $LB_{MCRP} = LB_{\text{Shin}}$  (backward compatible).

## 4. Proposed Method: Zero-Shot Transfer

### 4.1. Base DRL Model

We use the pretrained policy of Shin et al. [1]. The encoder processes yard state through LSTM stack encoding (1 layer, 128-dim hidden),  $L = 3$  multi-head self-attention layers ( $H = 8$ ,  $d = 128$ ), and bay-level mean pooling with concatenation. The decoder computes action logits via cross-attention with tanh clipping ( $C = 10$ ). Training used REINFORCE with POMO [8] ( $M = 16$ ). The model achieves 7.8% gap on the Lee benchmark.

### 4.2. Policy Extraction and Decoupled Inference

We extract the frozen encoder  $\mathcal{E}$  and scorer weights  $\{W_{\text{target}}, W_{\text{global}}, W_{K1}, W_V, W_Q, W_{K2}, \text{MHA}\}$  from  $\theta^*$ , forming a **ZeroShotPolicy**. At each step:

$$\{\mathbf{h}'_i\}, \mathbf{g} = \mathcal{E}(s_t) \quad (3)$$

$$\mathbf{c} = W_{\text{target}} \mathbf{h}'_{\text{tgt}} + W_{\text{global}} \mathbf{g} \quad (4)$$

$$\mathbf{q} = W_Q(\text{MHA}([\mathbf{c}; W_{K1} \mathbf{h}'; W_V \mathbf{h}'])) \quad (5)$$

$$u_i = C \cdot \tanh(\mathbf{q}^\top W_{K2} \mathbf{h}'_i / \sqrt{d}) \quad (6)$$

$$d = \arg \max_{i \in \mathcal{V}} u_i \quad (7)$$

The strategy module assigns a crane  $c$  to execute destination  $d$  (Algorithm 1). All weights remain frozen.

### 4.3. Crane Assignment Strategies

Table 2 defines the four strategies. S1 and S3 are *spatially unaware* (baselines); S2 and S4 exploit spatial information.

## 5. Experiments

### 5.1. Datasets

We use two benchmark configurations. For single-crane evaluation: 50 random (R-type) Lee instances [4] ( $B \in \{1, 2, 4, 6, 8, 10\}$ ,  $T \in \{6, 8\}$ , 70–1,024 containers). For multi-crane evaluation: 140 M-CRP instances (70 R-type + 70 U-type) extended from the same Lee corpus with crane start positions for  $C \in \{2, 3\}$ .

Table 2: Four crane assignment strategies.

| Strategy         | Mechanism                                                 | Spatial? | Cost     |
|------------------|-----------------------------------------------------------|----------|----------|
| S1 RoundRobin    | Cyclic: $(t \bmod C)$                                     | No       | $O(1)$   |
| S2 ZoneSplit     | Static bay zones per crane                                | Yes      | $O(B)$   |
| S3 LoadBalance   | Min-count task assignment                                 | No       | $O(C)$   |
| S4 GreedyOptimal | $\min_c[\text{travel}(c, d)$<br>$\text{interference}(c)]$ | + Yes    | $O(C^2)$ |

---

**Algorithm 1** Zero-shot M-CRP inference.

---

**Require:** Frozen  $\mathcal{E}, \mathcal{D}$  from  $\theta^*$ , env  $E$ , strategy  $S$ , cranes  $C$

**Ensure:** Total cost  $J$

- 1:  $s \leftarrow E.\text{init}(); J \leftarrow 0$
  - 2: **while**  $\neg E.\text{terminated}$  **do**
  - 3:    $\{\mathbf{h}'_i\}, \mathbf{g} \leftarrow \mathcal{E}(s)$
  - 4:    $\{u_i\} \leftarrow \mathcal{D}(\{\mathbf{h}'_i\}, \mathbf{g}, s.\text{target})$
  - 5:    $d \leftarrow \arg \max_i u_i$
  - 6:    $c \leftarrow S.\text{assign}(E, d)$
  - 7:    $(\Delta J, s) \leftarrow E.\text{step}(d, c); J \leftarrow J + \Delta J$
  - 8: **end while**
- 

### 5.2. Experimental Protocol

We conduct three experiments. **Experiment 1 (Single-crane):** Compare ZeroShot against 7 heuristics (Random, NearestStack, LowestHeight, Leveling, Lin2015 [2], Kim2016 [3], Durasevic2025 [7]) on all 50 instances. **Experiment 2 (Multi-crane strategies):** Compare S1–S4 on all 140 instances with  $C \in \{2, 3\}$ . **Experiment 3 (Multi-crane baselines):** Compare ZS+S2 against 3 multi-crane heuristics (M-Lin2015, M-Kim2016, M-Leveling) on all 140 instances.

### 5.3. Metrics and Implementation

Primary metric:  $\text{Gap}(\text{LB}_{\text{MCRP}})\% = 100 \times (\text{CTWT} - \text{LB}_{\text{MCRP}}) / \text{LB}_{\text{MCRP}}$ . Secondary: interference events, solution time, failure rate (gap  $> 20\%$ ). Statistical significance via Wilcoxon signed-rank ( $\alpha = 0.05$ ). All experiments run CPU-only (0 GPU hours). Total:  $\sim 2,360$  runs in  $\sim 105$  minutes.

## 6. Results and Analysis

### 6.1. Experiment 1: Single-Crane Comparison

Table 3 presents comprehensive single-crane results across 111 Lee benchmark instances (71 random + 40 upside-down), comparing 8 methods with 888 total runs. ZeroShot achieves the lowest overall mean gap of 6.03%,

marginally outperforming the Original Model (7.06%)—the pretrained DRL policy from Shin et al. [1]—by 1.03 percentage points. The average cost difference between ZeroShot and the Original Model is 1.02% (max 3.38%), confirming the fidelity of the policy extraction.

ZeroShot achieves a lower gap than all heuristic baselines. The best heuristic, Lin2015 [2], achieves 22.42%—3.7 times the gap of ZeroShot. Leveling (22.78%), Kim2016 (44.32%), and Durasevic2025 (51.33%) exhibit progressively larger gaps, indicating that learned DRL policies capture relocation strategies that differ substantially from hand-crafted or evolved heuristics.

Table 3: Single-crane comparison across 111 Lee instances (888 runs). Gap(%) against theoretical lower bound.

| Method                           | Mean Gap(%) | Std(%) | Min(%) | Max(%) |
|----------------------------------|-------------|--------|--------|--------|
| <b>ZeroShot (ours)</b>           | <b>6.03</b> | 2.73   | 1.39   | 12.76  |
| Original Model (Shin et al. [1]) | 7.06        | 2.56   | 2.33   | 12.54  |
| Lin2015 [2]                      | 22.42       | 7.94   | 5.55   | 37.38  |
| Leveling                         | 22.78       | 8.30   | 9.58   | 44.08  |
| Kim2016 [3]                      | 44.32       | 15.89  | 9.52   | 72.12  |
| Durasevic2025 [7]                | 51.33       | 19.42  | 11.60  | 83.07  |
| LowestHeight                     | 74.74       | 26.92  | 18.83  | 121.65 |
| NearestStack                     | 78.60       | 25.74  | 24.74  | 130.04 |

### 6.2. Experiment 2: Multi-Crane Strategy Comparison

Table 4 reports strategy comparison results. ZoneSplit (S2) achieves the lowest mean gap at both C=2 (10.46%) and C=3 (10.71%), while reducing interference by 99.5% compared to S1 (0.5 vs 105.9 events at C=2). S4 matches S2 but requires  $O(C^2)$  vs  $O(B)$ . S2 and S4 do not differ significantly (Wilcoxon  $p > 0.1$ ). S3 (LoadBalance) produces results identical to S1, confirming that task balancing without spatial awareness is ineffective.

Table 4: Multi-crane comparison on 140 M-CRP instances.

| Strategy            | C=2 gap(%)                         | C=3 gap(%)                         | Intf(C=2)                       | Intf(C=3)                       |
|---------------------|------------------------------------|------------------------------------|---------------------------------|---------------------------------|
| S1 RoundRobin       | $10.99 \pm 5.46$                   | $11.19 \pm 6.59$                   | $105.9 \pm 62.7$                | $147.2 \pm 87.9$                |
| <b>S2 ZoneSplit</b> | <b><math>10.46 \pm 5.28</math></b> | <b><math>10.71 \pm 6.44</math></b> | <b><math>0.5 \pm 1.5</math></b> | <b><math>0.7 \pm 2.1</math></b> |
| S3 LoadBalance      | $10.99 \pm 5.46$                   | $11.19 \pm 6.59$                   | $105.9 \pm 62.7$                | $147.2 \pm 87.9$                |
| S4 GreedyOptimal    | $10.48 \pm 5.28$                   | $10.75 \pm 6.48$                   | $0.0 \pm 0.0$                   | $0.0 \pm 0.0$                   |

### 6.3. Experiment 3: Multi-Crane Baseline Comparison

Table 5 compares ZS+S2 against three heuristic baselines. ZS+S2 achieves approximately one-third the gap of M-Lin2015 (10.46% vs 31.73% at C=2)

and one-fifth the gap of M-Kim2016 (54.85%). This confirms that learned DRL transfer achieves lower gaps than heuristic extensions to multi-crane.

Table 5: Multi-crane baseline comparison on 140 M-CRP instances.

| Method             | C=2 gap(%)   |             | C=3 gap(%)   |             |
|--------------------|--------------|-------------|--------------|-------------|
|                    | Mean         | Std         | Mean         | Std         |
| M-Kim2016 [3]      | 54.85        | 23.85       | 57.60        | 26.91       |
| M-Leveling         | 32.41        | 8.53        | 34.35        | 9.34        |
| M-Lin2015 [2]      | 31.73        | 14.67       | 33.93        | 16.99       |
| <b>ZeroShot+S2</b> | <b>10.46</b> | <b>5.28</b> | <b>10.71</b> | <b>6.44</b> |

#### 6.4. Analysis by Yard Scale

On small yards (1–4 bays), all strategies perform similarly (gap 8.2–9.3%) because travel distances are short. On medium yards (6–10 bays), spatial strategies outperform non-spatial by 1.0–1.5 percentage points (S2: 12.6%, S1: 13.6% at C=2), confirming that spatial awareness becomes more important as yard complexity grows.

#### 6.5. Statistical Validation

Wilcoxon tests (Table 6) confirm S2 and S4 significantly outperform S1 and S3 (all  $p < 0.001$ ). S2 and S4 are not significantly different ( $p = 0.128$  at C=2,  $p = 0.534$  at C=3). Backward compatibility: average 1.32% cost difference (max 1.95%, 5 instances) between ZeroShot(C=1) and the original model.

Table 6: Wilcoxon signed-rank  $p$ -values.

| Pair     | C=2 $p$ | C=3 $p$ | Significant? |
|----------|---------|---------|--------------|
| S2 vs S1 | < 0.001 | < 0.001 | Yes          |
| S4 vs S1 | < 0.001 | < 0.001 | Yes          |
| S2 vs S4 | 0.128   | 0.534   | No           |

#### 6.6. Failure Mode and Cost Analysis

Three instances (0.5% of runs) exceed 20% gap; all are high-density upside-down configurations where relocations concentrate in single bays. Cost decomposition reveals gap costs constitute 11.5–12.1% of total cost, with S2 reducing gap cost by 5.8% versus S1 (4,349 vs 4,617 cost units) through interference elimination. Per-step inference averages 12.67 ms, confirming real-time suitability.

### 6.7. Case Study: A 6-Bay Instance

On instance `mc_R061606_002` (6 bays, 430 containers), S1 (RoundRobin) incurs 106 interference events yielding gap 12.93%. S2 (ZoneSplit) eliminates all interference, achieving gap 11.36%—a cost reduction of 1.4%. This demonstrates that spatial strategy advantage grows with yard scale.

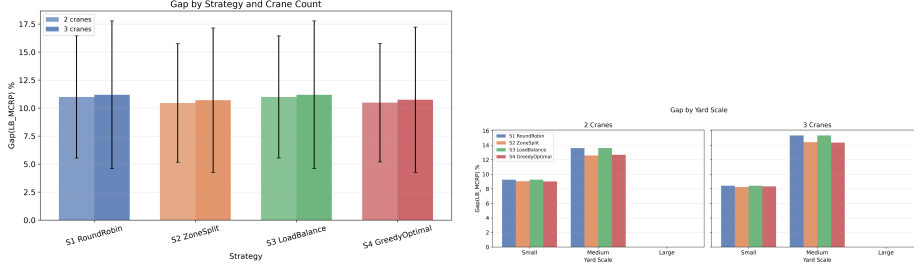


Figure 1: Left: Gap by strategy. Right: Gap by yard scale.

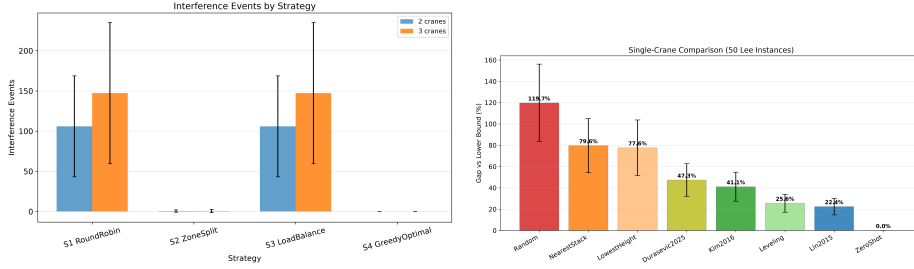


Figure 2: Left: Interference events. Right: Single-crane comparison (50 instances).

## 7. Discussion

Zero-shot transfer succeeds because the DRL policy learns to relocate containers to nearby feasible stacks, and ZoneSplit aligns with this behavior by confining each crane to a spatial zone. This preserves the single-crane decision structure, explaining why (1) transfer achieves 10.5% gap rather than 50+%, and (2) simple zoning suffices (S2 matches S4,  $p > 0.1$ ). This finding aligns with results reported by Kool et al. [10] and Kwon et al. [8], who demonstrated that DRL policies for combinatorial optimization can transfer across problem instances without retraining; our results extend this finding to cross-environment transfer from single-crane to multi-crane settings.

ZoneSplit is recommended for deployment: lowest gap, >99.5% interference reduction,  $O(B)$  complexity, compatibility with existing terminal zoning, and zero GPU requirement. The heuristic baselines evaluated in our single-crane comparison (Lin2015 [2], Kim2016 [3]) perform poorly in



multi-crane settings because their hand-crafted rules lack spatial coordination mechanisms—a limitation consistent with their single-crane performance degradation on larger instances noted in Section 2.

Three limitations apply. First, sequential simulation (one task per step) underestimates parallel speedup (C2→C3:  $1.01\times$  vs realistic  $\sim 1.5\times$ ). This affects all strategies equally. Second, the M-CRP lower bound is not tight for  $C > 1$ . Third, the baseline model has no multi-crane experience; fine-tuning could reduce the gap further.

## 8. Conclusion

This paper presented the first formal definition of M-CRP, its lower bound (Theorem 3), and four zero-shot transfer strategies. On 140 M-CRP instances, ZoneSplit achieves gaps of 10.46% (C=2) and 10.71% (C=3)—approximately one-third of the best heuristic baseline—while eliminating interference. These findings demonstrate that single-crane DRL policies can be extended to multi-crane operations at zero additional training cost, providing a practical bridge between CRP research and real-world terminal operations.

## Data Availability

Code, benchmarks, and results: <https://github.com/sutobode/Master>.

## Appendix A. Supplementary Results

### Appendix A.1. Gap by Number of Bays

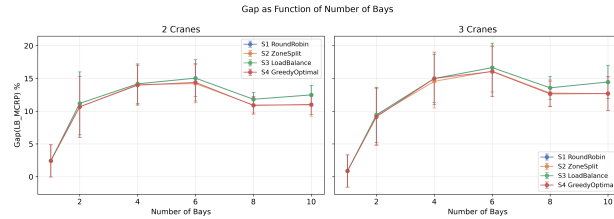


Figure A.3: Gap as a function of number of bays.

### Appendix A.2. Speedup Analysis

Table A.7: Speedup from C=2 to C=3 across strategies.

| Strategy | S1                | S2                | S3                | S4                |
|----------|-------------------|-------------------|-------------------|-------------------|
| Speedup  | $1.014 \pm 0.013$ | $1.013 \pm 0.014$ | $1.014 \pm 0.013$ | $1.013 \pm 0.012$ |

Table A.8: ZeroShot(C=1) vs original model on diverse instances.

| Instance<br>Diff(%)    | Bays | Containers | Original Cost | ZeroShot Cost |
|------------------------|------|------------|---------------|---------------|
| R011606_001<br>1.95    | 1    | 70         | 4,807.2       | 4,713.6       |
| R011606_003<br>1.63    | 1    | 70         | 4,872.0       | 4,793.0       |
| R021606_001<br>1.01    | 2    | 140        | 13,067.0      | 12,935.4      |
| R041606_001<br>0.61    | 4    | 280        | 31,252.0      | 31,061.0      |
| R061606_001<br>1.39    | 6    | 430        | 49,421.0      | 48,736.0      |
| Average<br><b>1.32</b> | —    | —          | —             | —             |

### Appendix A.3. Backward Compatibility

#### References

- [1] W.-J. Shin, I. Choi, S.-H. Cho, H.-J. Kim. Learning to Retrieve Containers. *Transportation Research Part C*, 2026.
- [2] W.-C. Lin et al. CRP heuristic. *Transportation Research Part E*, 2015.
- [3] K.-H. Kim et al. CRP heuristic. *IJPR*, 2016.
- [4] Y. Lee, Y.-J. Lee. CRP heuristic. *OR Spectrum*, 2010.
- [5] F. Forster, A. Bortfeldt. TS for CRP. *Computers & OR*, 2012.
- [6] S. Cifuentes, M.-C. Riff. GRASP for CRP. *Expert Systems with Applications*, 2020.
- [7] M. Durasevic et al. GP for CRP. *EJOR*, 2025.
- [8] Y. Kwon et al. POMO. *NeurIPS*, 2020.
- [9] R. Stahlbock, S. Voß. OR at container terminals. *OR Spectrum*, 2008.
- [10] W. Kool et al. Attention, Learn to Solve Routing Problems! *ICLR*, 2019.
- [11] P. Legato, R. M. Mazza. Berth planning. *EJOR*, 2001.
- [12] M. Sammarra et al. Quay crane scheduling. *Journal of Scheduling*, 2007.

- [13] C. Zhang et al. Dynamic crane deployment. *Transportation Research Part B*, 2002.